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Data compression using SVD and Fisher information for radar emitter location

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ABSTRACT

This paper presents a method that achieves large compression ratios for radar pulse trains exploited in time-difference-of-arrival/frequency-difference-of-arrival (TDOA/FDOA) emitter location; this method exploits pulse-to-pulse redundancy to get a compression ratio much higher than possible by using standard methods. We show how to use (i) the singular value decomposition (SVD) to exploit redundancy between radar pulses, and (ii) a Fisher information-based distortion criterion to enable elimination of pulses that are negligible to the FDOA estimation task. To use the SVD it is necessary to first gate the pulses, place them in a matrix and then align the pulses to get a matrix that is close to rank one. Finally, we employ reasonable coding schemes for the information to be sent and assess the achievable compression level. With such SVD-based coding the compression ratio depends on the number of pulses intercepted and the number of samples taken per pulse; depending on the scenario the compression ratio can range from about 4 up to 100 or more.

Fisher information-based removal of pulses is shown to improve the compression ratio beyond what is achievable using only the SVD approach while having negligible impact on the FDOA accuracy; it does, however, degrade the TDOA accuracy. However, we demonstrate that the SVD method includes an inherent de-noising effect (common in SVD-based signal processing) that provides an improvement in TDOA accuracy over the case of no compression processing; thus, the overall impact on TDOA/FDOA accuracy can be negligible while providing significant compression ratios for typical radar signals.

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1. Introduction

A common way to locate electromagnetic emitters is to measure the time-difference-of-arrival (TDOA) and the frequency-difference-of-arrival (FDOA) between pairs of signals received at geographically separated platforms [1,2]. The measurement of TDOA/FDOA between these signals is done by coherently cross-correlating the signal pairs [3,4]. This requires that the signal samples of the two signals are available at a common platform, which is accomplished by transferring the signal samples over a data link from one platform to the other. It should be noted the TDOA/FDOA location problem has applications in satellites and airborne systems such as SARSAT and EUTELSAT [5,6]. Also, although Refs. [1–4] provide a solid basis for TDOA/FDOA-based location processing, there has been continuing work in this area up to the present time; there are too many relevant references to list but two recent developments can be found in [7,8]. There are also many results for TDOA/FDOA that come from the sonar processing field (one example is by Weinstein [9]); however, has been pointed out in [10], there are some important distinctions in TDOA/FDOA results for sonar location vs. RF location.

An important aspect of this processing that was not widely addressed in the literature until recently is that the available data link rate is often insufficient to accomplish

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the transfer within the time requirement unless some form of lossy data compression is employed. To mitigate this, we have developed data compression methods tailored specifically for TDOA/FDOA emitter location systems. In [11–13] we showed how to compress radar signals by "gating" around the detected pulses, putting the gated pulses into the rows of a "pulse matrix", and then using the singular value decomposition (SVD) to compress the signal by exploiting redundancy between pulses when the emitter to be located is a radar; this approach employed a purely MSE distortion criterion. Alternatively, in [14–16] we have developed a method that uses Fisher information as a distortion measure that captures the true impact of compression on TDOA/FDOA accuracy; this has been applied in a very general way to signals that are pulsed or non-pulsed to determine optimal allocation of bits to time-frequency components of the signal to be compressed.

In this paper we start with our pulse matrix approach and extend it by specifically applying the Fisher information-based distortion measure for FDOA accuracy to find the optimal set of pulses to remove from the pulse matrix and then use the SVD for transform-based compression. The removal of pulses is shown to have negligible impact on the FDOA accuracy but does degrade the TDOA accuracy from that achievable using the SVD-based compression without pulse elimination. However, we demonstrate that the SVD method includes an inherent de-noising effect (common in SVD-based signal processing) that provides an improvement in TDOA accuracy over the case of no compression processing; thus, the overall impact on TDOA/FDOA accuracy is negligible while providing compression ratios that can range from about 4 up to 100 or more for typical radar signals.

The method presented here is aimed at compression within systems that passively intercept signals from a non-cooperative radar. However, we point out that compression methods do exist for use within cooperative radar systems, but those methods are very different from what we propose here because of the different signal models and processing environments. For example, compression methods exist for synthetic aperture radar (SAR) [17–19] where, unlike in the TDOA/FDOA geo-location scenario, the signal to be compressed is modeled as a Gaussian signal in additive Gaussian noise. In SAR the Gaussian signal model is relevant (in light of the central limit theorem) due to the fact that the received signal consists of a large number of ground reflections, whereas in TDOA/FDOA of RF signals a Gaussian signal model is not the accepted one [3]. Another example is compression in a cooperative network of radars, where the compression would operate on the signal data that occurs *after* the correlation processing of the matched filter, whereas in the TDOA/FDOA case the compression occurs *before* cross-correlation processing (there is no matched filter available due to the non-cooperative setting). Thus, compression algorithms for SAR and other cooperative radar applications will take a very different nature than those for TDOA/FDOA scenarios. Finally, there have been a few other methods proposed for non-cooperative TDOA/FDOA scenarios (as well as other non-cooperative scenarios that do not perform TDOA/FDOA, such as sensor networks) and those are summarized in the introduction of [15]. The method presented here is unique in that it jointly exploits pulse-to-pulse redundancy and the Fisher information-based distortion criteria we have used in more general TDOA/FDOA settings.

In Section 2 we provide an overview of the compression method. In Section 3 we provide the details of the method, in Section 3.1 we describe an optimal method for gating the pulses and placing them into the pulse matrix so as to minimize the amount of side information that needs to be sent; in Section 3.2 we describe how to align the pulses to reduce the rank of the matrix; in Section 3.3 we show how to use the SVD to extract a prototype pulse and a set of complex amplitudes that allow reconstruction of the pulse train; in Section 3.4 we provide an analysis of the achievable compression ratio; in Section 3.5 we show how to use Fisher information to eliminate pulses that are negligible for the FDOA estimation. Section 4 provides illustrative simulation results and Section 5 provides a brief conclusion.

2. Background and overview

2.1. Emitter location system background

Before discussing our compression method, in this subsection, we briefly describe the system-level scenario we assume for processing pulsed radar signals in a TDOA/FDOA-based geo-location system; this is the environment in which our compression method must work. This is the particular scenario of our interest, although there are other related scenarios that can be handled in a similar way.

The emitter location system consists of three or more platforms, each outfitted with identical receiving and processing equipment. Once signal data from the non-cooperative emitter-of-interest is intercepted at all of the platforms, the SNR is estimated at each platform and the one with the highest SNR is chosen as the one to transmit its data to the others for subsequent correlation processing; this platform is called the transmitting platform (Tx platform) and the other platforms are called the correlating platforms (Cx platforms). The Tx platform is required to be at a high enough SNR to allow standard radar intercept signal processing to be done [20]. Because modern radars can change modes we assume that preliminary sub-train-extraction (de-interleaving) processing has grouped the signal of interest into one or more sub-trains, each having pulses from the same mode of operation—such processing is a standard part of typical electronic warfare systems (this processing also removes pulses from other emitters) [20]. Here we consider compressing one such sub-train.

As part of this interception processing, the Tx platform detects the individual pulses of the emitter-of-interest, gates around them, and keeps only the signal samples that lie inside the pulse gates. The gated signal samples and numbers of removed samples are transmitted to each of the Cx platforms where this broadcast data is used to reconstruct the gated pulse train (zeros are inserted between pulses according to the side information).

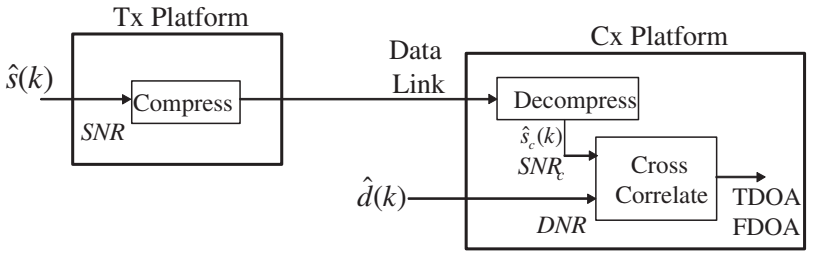


Fig. 1. System configuration for compression.

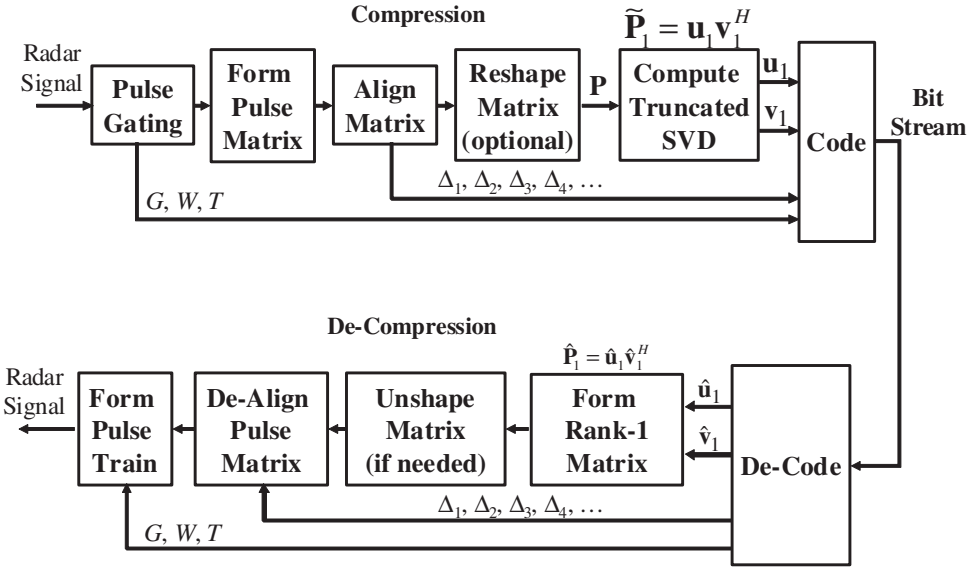


Fig. 2. Overview of compression and de-compression method.

At each Cx platform the reconstructed Tx platform signal is then cross-correlated with the Cx platform's locally received signal to give ML estimation (via cross-correlation) of the TDOA/FDOA values [3,4]. The sets of TDOA/FDOA estimates determined at the various Cx platforms are then combined to estimate the emitter location [1,2]. We focus on the compression of the data transferred from Tx to Cx as shown in Fig. 1.

The two noisy intercepted signals are modeled as

$$\hat{s}(k) = s(k) + n(k) \quad \hat{d}(k) = d(k) + v(k), \tag{1}$$

where $s(k)$ and $d(k)$ are the discrete-time complex baseband signals of interest and $n(k)$ and $v(k)$ are zero-mean complex white Gaussian noises that have the same variance and are uncorrelated with each other. The continuous-time signal $d(t)$ that gives rise to $d[k]$ is an amplitude-scaled, delayed and doppler shifted version of the continuous-time signal $s(t)$ that gives rise to $s[k]$: $d(t) = \alpha s(t-D)e^{j\omega_d t}$. The signal-to-noise ratios (SNR) for these two signals are denoted *SNR* and *DNR*, respectively[1].

[1] SNR (non-italic) represents an acronym for signal-to-noise ratio; *SNR* (italic) represents the SNR for $\hat{s}(k)$ and *DNR* (italic) represents the SNR for $\hat{d}(k)$.

After lossy compression/decompression of $\hat{s}(k)$ the resulting signal $\hat{s}_c(k)$ has SNR of SNR_c.

2.2. Overview of compression method

The data compression/de-compression processing is outlined in Fig. 2. The first two steps in the data compression are to refine the pulse gating done by the standard radar interception processing to get equal length extracted pulses and then form the pulse matrix; (see Fig. 3). To increase the amount of compression the pulse gating method optimally chooses the integers *G*, *W* and *T* in units of "samples" (see Section 3.1), the values of which are sent as side information. Each gated pulse is put into a row of the pulse matrix and then the pulses in the rows are aligned to give the aligned pulse matrix (see Section 3.2). The amount of alignment imparted to each pulse is sent as side information in the sequence Δ_1, Δ_2, Δ_3,... as shown in Fig. 2. If desired, the aligned pulse matrix can be reshaped by removing pulses deemed to be negligible via a Fisher information-based distortion measure (see Section 3.5). The aligned pulse matrix (with or without pulse removal) is close to being rank one and the SVD is used to extract the most significant left

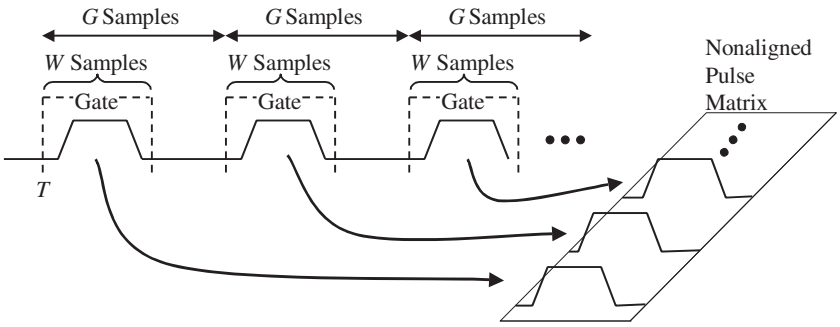


Fig. 3. Pulse gating and pulse matrix formation.

and right singular vectors, $\mathbf{u}_1$ and $\mathbf{v}_1$ (see Section 3.3). The extracted SVD vectors and all the side information are then efficiently coded (see Section 3.4) and transmitted to the Cx platform where the process is reversed to reform an approximation to the original pulse train.

3. Details of compression method

3.1. Pulse gating and pulse matrix formation

Given starting times $\{t_k\}$ and widths $\{w_k\}$, $k=1, 2, \ldots, p$, for the p pulses intercepted by the front-end radar interception processing, the pulse gating step determines refines gates that are defined by the three parameters T, G and W shown in Fig. 3; T is the "gating origin", G is the gate interval, and W is the gate width. This pulse gating step is done so as to accomplish two goals: (i) describe the gating via three constants rather than the sequence of t_k and w_k values from the front-end and (ii) ensure that the resulting gate width W is a minimum. The former reduces the amount of side information (send only T, G, and W rather than sending all the t_k and w_k values); the latter ensures that the width of the resulting pulse matrix is minimized (which reduces the amount of signal data that needs to be sent). Once the gating is determined it is used to gate the pulses and put one gated pulse per row of the pulse matrix.

To formulate the pulse gating problem, recall that we wish to minimize the gating pulse width W, which leads to specifying the following integer optimization problem:

$$\begin{aligned}
&\text{Given}: \ \{t_k, w_k\}, \quad \text{for } k=\{0,1,\ \ldots, p-1\}\\
&\text{Choose}: \ T, G>0, W>0 \quad \text{(integers)}\\
&\text{To minimize}: \ W\\
&\text{While satisfying}:\\
&kG+T \le t_k, \quad k=\{0,1,\ \ldots, p-1\}\\
&kG+T+W \ge t_k+w_k, \quad k=\{0,1,\ \ldots, p-1\}.
\end{aligned} \tag{2}$$

From this formulation we see that since the objective function to be minimized is W; this obviously is linear in W and there are constraints on W, G, and T—thus, we have cast this problem as a traditional linear optimization problem with inequality constraints [21]. Although there are standard methods to solve this we develop a problem-specific method that has complexity advantages over the standard methods.

To get a better perspective of this minimization problem, Fig. 4 shows t_k and (t_k+w_k) vs. k along with two lines of slope G that bound these two sets of points. On the left side of the figure the corresponding pulse gates are represented by shaded rectangles. Thus, the minimization problem to be solved is: pick integers T, $G>0$, $W>0$ so that the two lines enclose the two sets of points while minimizing the vertical distance between the lines.

We start the development of our optimization method by fixing G to some initial value, a positive integer. From the perspective of Fig. 4, the line $t=kG+T$ must pass though at least one of the leading edge points. If it did not, then W could be reduced by incrementing T until the line did pass through one of the leading edge points. From this, we find that if G is fixed, then T can be chosen such that this condition holds. In fact, it leads to this equation for selecting T, given G

$$T=\min_k[t_k-kG]. \tag{3}$$

A similar argument shows that for fixed G and T, W can be optimally chosen as

$$W=\max_k[(t_k+w_k)-(kG+T)]. \tag{4}$$

Note that in Eqs. (3) and (4), if G is an integer, then so are T and W (since t_k and w_k are integers). Now it is a matter of finding the value of G that minimizes W.

It can be easily seen that the optimal W as a function of G is unimodal—where T is considered to take on its optimal value for each G as defined in Eq. (3). This arises from Eq. (2) as follows. Imagine the three dimensional space with axes W, G, and T. The first constraint in Eq. (2) specifies a series of vertical planes (i.e., parallel to the W-axis) and Fig. 5 shows the intersection of these constraint planes with the G–T plane; the shaded area identifies the region satisfying these constraints. From our previous considerations we know that the solution must lie on one of the lines forming the constraint region. The second constraint in Eq. (2) places a series of tilted planes above the G–T plane, each of which has a slope of -1 with respect to the T-axis and the kth plane has a slope of $-k$ with respect to the G-axis. The conglomeration of these constraint planes creates a convex downward surface. Thus, W as a function of G has the values on this convex downward surface as traversed along the conglomeration of the constraint lines in Fig. 5, which creates a unimodal

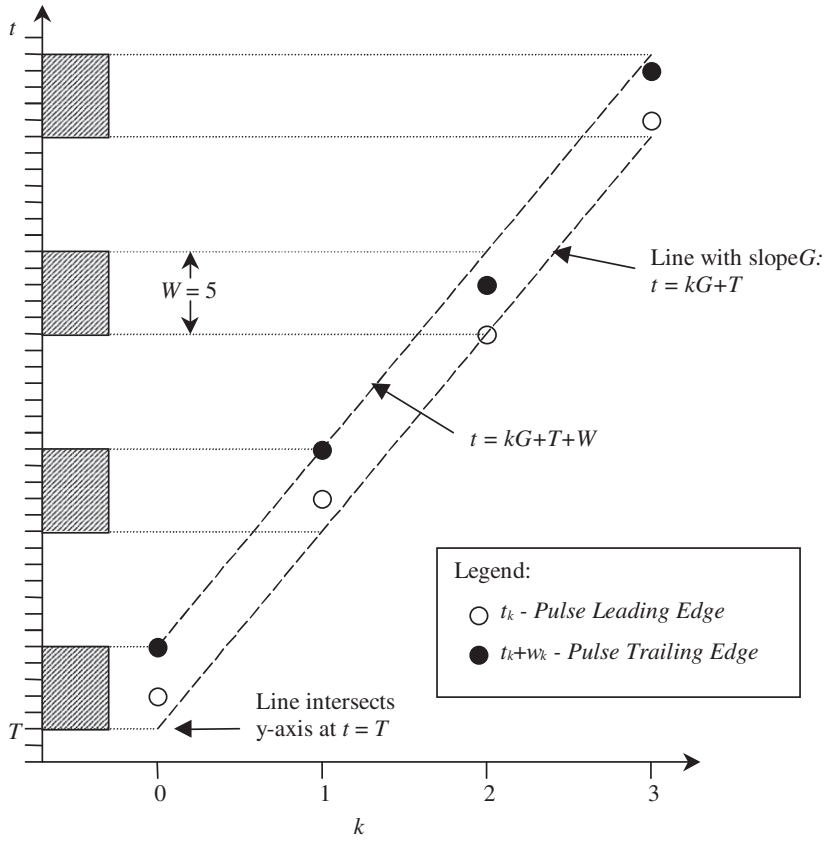


Fig. 4. Pulse leading and trailing edge times vs. pulse index *k*.

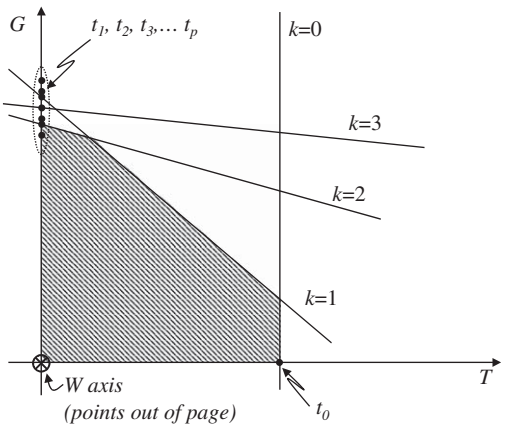


Fig. 5. Example of constraints in the *T*–*G* plane.

function with a single minimum. A property of a unimodal function is that if a local minimum is located, it will be the global minimum. Various well-known algorithms (such as the bisection method, golden section search, etc.) can now be used to solve the optimization problem.

There are several different methods that can be used to select the initial trial value needed here for *G*. The non-integer simplex method may be used. This tends to locate the optimal value for *G* within a fraction, but the method is complicated (this method also places a lower bound on *W*). A least-square fit to the slope of the leading edge sample points may be used to estimate *G*. The method is simple, but may require more trials to reach the optimal *G*. A simple slope calculation between the leading edges of the first and last pulse may also be useful.

Our optimal "unimodal method" thus consists of finding an initial estimate for *G* (call it G_1) using any one of the methods mentioned in the previous paragraph and then updating to $G_2=G_1+1$, solving Eqs. (3) and (4) for each of G_1 and G_2 to get W_1 and W_2, and then from this information the unimodality can be used to determine the direction to the minimum of $W(G)$; *G* is then incremented in unit steps until $W(G)$ – checked using Eqs. (3) and (4) – starts to increase, at which point the optimal *G*, *T*, and *W* have been found. The efficiency of this method depends on the accuracy of the initial estimate of *G*, but it executes rapidly and its run time grows very slowly with the number of pulses. In particular, we have shown (see [12] for details) that our "unimodal" method is at least 2 orders of magnitude faster than the standard optimization methods [21,22] such as simplex, simplex with convex hull, dual simplex, dual simplex with convex hull, and dual simplex with convex hull and cutting planes; also, the run time of our method grows much slower with increasing number of pulses (it is nearly flat) than for the standard methods.

An implicit assumption has been made in the above discussion of the gating: there are no missing pulses (i.e., undetected due to low SNR, for example). The occurrence of a missing pulse can be recognized from irregular spacing between leading edge times and is a standard task in radar intercept processing. Although missing pulses would have no impact on the algorithm to find the optimal W, G, and T nor on the subsequent pulse matrix alignment, the impact of this on the amount of side information to be sent must be addressed. We propose here to use a simple bit mask to indicate where there are missing pulses: the bit mask would have a "1" to indicate a pulse is present and a "0" to indicate that a pulse is missing; there are better ways to do this exploiting run-length coding and/or other methods of entropy coding but there use is quite standard and we do not consider them here. Thus, for a given number of intercepted pulses p, the length of this bit mask will depend on the probability of a missing pulse P_{mp}. If there are p intercepted pulses, then the expected number of transmitted pulses (and hence the length of the bit mask) is $p/(1-P_{mp})$ bits. Thus, if we consider a somewhat worst case scenario, with a value of say P_{mp}=0.5, then the expected number of bits in the bit mask would be $2p$ bits. If no missed pulses are identified then this bit mask could be eliminated and that condition could be flagged in the small amount of header information used in all practical compression methods.

3.2. Pulse matrix alignment

Once the optimal gating has been determined the pulse matrix is formed by putting each optimally gated pulse into a row of a matrix; but there is no guarantee that the pulses in these rows are well-aligned. Let $\mathbf{P}_{na}$ be this matrix whose rows hold the non-aligned pulses that are extracted by the gating procedure, as shown in Fig. 3; it has p rows (where p is the number of intercepted pulses) and has W columns (where W is the optimal gate width found in Section 3.1). The goal now is to align the pulses in the rows to yield an "aligned" matrix $\mathbf{P}$ that is as close to rank one as possible to enable compression via the SVD. Because the radar's PRI and the platform's sampling interval T_s are generally incommensurate (i.e., there is no *integer* k such that PRI=kT_s) the needed time alignment is not an integer number of samples. Therefore, we need a method of shifting pulses by a fraction of a sampling interval.

Let $p_j(k)$ be the jth pulse so that matrix $\mathbf{P}_{na}$ has its j,k element given by $\mathbf{P}_{na}(j,k)=p_j(k)$. We choose the pulse with the largest energy as the reference pulse, to which all the other pulses will be aligned; let this pulse be denoted as $p_m(k)$. Then to find the time alignment needed for the jth pulse ($j\neq m$) we cross-correlate it with the reference pulse to give

$$C_j(k)=\sum_l p_m(l)p_j^*(l-k), \quad j\neq m, \tag{5}$$

and then interpolate $C_j(k)$ to find the location of its peak, which is then taken as the time shift Δ_j to be applied to $p_j(k)$ to align it with $p_m(k)$. The time shift Δ_j can be written as $\Delta_j=D_j+\delta_j$ where D_j is an integer and $0\leq\delta_j<1$.

The integer part of the alignment can be handled separately according to $\tilde{p}_j(k)=p_j(k+D_j)$. Now $\tilde{p}_j(k)$ must be shifted by an amount that is less than a single sample. There are several ways to impart a fractional delay to a signal [11,13,23]. Which one should be used depends partly on whether the signal to be delayed is available in its entirety as one block or is available sequentially, either sample-by-sample or on a subblock-by-subblock basis; that is, it depends on the level of latency that is acceptable. Other considerations are accuracy and complexity. Here we consider four different methods and assess them on these merits. The methods are (i) a full-block DFT method based on the delay property of the DFT [13], (ii) a subblock-based version of the DFT method [13], (iii) fixed FIR filters designed for fractional delay [23], and an adaptive FIR filter for fractional delay [13]. Zhou [11,13] compared the accuracy on the basis of magnitude vs. frequency, delay vs. frequency, and SNR vs. delay value. Because in this application the signals to be delayed (the individual pulses) can be quite short, the adaptive FIR method is not suitable. The fixed FIR methods are also not the best choice because (i) the signals can be short, (ii) the signals are wideband so the frequency-specific characteristic is a major disadvantage, and (iii) the worst-case accuracy vs. delay may not be acceptable. That leaves the subblock DFT and the full-block DFT methods. For the current application there is no penalty in having to wait for an entire pulse to become available because the front-end processing must first identify all the pulses before any subsequent processing is done; furthermore, the required fractional delay cannot be determined until the entire pulse is available. Therefore, for this application we use the full-block DFT approach [13].

The effectiveness of the fractional delay method for this application can be seen by assessing its ability to reduce the effective rank of the pulse matrix. The effective rank of a matrix is easily assessed via the SVD [24]. If the alignment method is effective at creating a matrix with effective rank one, then all but the first singular value of the aligned matrix $\mathbf{P}$ should be insignificant. Because the sum of the squares of the singular values gives the energy of the signal, it makes more sense to plot the squares of the singular values; for each case, normalizing by the largest singular value improves the comparison between various cases (e.g., non-aligned, aligned, etc.). Fig. 6 shows the squares of the normalized singular values as a function of singular value index for the unaligned matrix, the aligned matrix using only integer alignment, and also using fractional alignment; the case shown here is for a simulated linear FM radar signal sampled at an interval that is incommensurate with the PRI. From this we see the effectiveness of the fractional alignment method—the effective rank of the fractionally aligned matrix can be seen to be close to one. This is the basis of the compression method developed here: compression is achieved because the fractionally aligned matrix can be closely approximated in terms of a rank one matrix.

3.3. SVD-based extraction

In the actual processing, the next step would reshape the aligned pulse matrix by removing negligible rows; although such pulse removal is optional it is typically

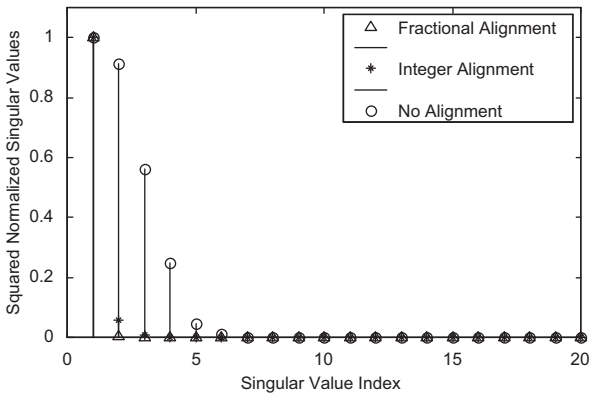


Fig. 6. Squared normalized singular values for aligned and unaligned matrices.

desirable to perform the reshaping step. However, we postpone discussion of the pulse removal step until later for ease of discussion; in addition, the results for compression ratio presented in Section 3.4 will motivate the reason for doing pulse removal. In either case—with or without the pulse removal—the idea behind the application of the SVD is the same.

Let the SVD of the $p \times W$ aligned pulse matrix be

$$\mathbf{P} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^H, \tag{6}$$

where p is the number of pulses, W is the number of gated samples per pulse, r is the rank of $\mathbf{P}$, $\mathbf{u}_i$ is the ith left singular vector, $\mathbf{v}_i^H$ is Hermitian transpose of the ith right singular vector, and σ_i is the ith singular value, ordered such that $\sigma_i \geq \sigma_{i+1}$. Each term in the sum in Eq. (6) is a rank-one matrix. In general, truncating this sum to only $k < r$ terms we get a rank-k matrix $\mathbf{P}_k$ that best approximates $\mathbf{P}$ in the sense that the sum of the squares of the elements of $\mathbf{P} - \mathbf{P}_k$ is smaller than for any other rank—k matrix [24]. Note that in our case the matrix contains the pulses and therefore this approximation gives the smallest mean square error (MSE) between the original pulse train and the approximate pulse train formed by concatenating the de-aligned rows of $\mathbf{P}_k$. MSE is a standard distortion measure for compression and it is this minimum MSE property that is the basis for using the SVD here. As illustrated in Fig. 6 the pulse alignment is expected to enable retaining only $k=1$ terms from Eq. (6) while maintaining a negligible MSE. Also, since the effect of the noise on the singular values is uniformly spread across all the singular values—this is in fact a known result that is exploited in many applications of the SVD to signal processing problems [25]—the SVD truncation reduces the noise. This simultaneous compression and noise reduction will be demonstrated in the simulations.

In particular, the approximating matrix $\mathbf{P}_1$ is formed as $\mathbf{P}_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^H$, from which it is clear that each row in $\mathbf{P}_1$ is a complex-valued scalar multiple of $\mathbf{v}_1^H$, where the complex scalar for the ith row is the ith element in $\mathbf{u}_1$ times σ_1; it is also clear that σ_1 does nothing more than amplitude scale the entire reconstructed pulse train and can therefore be omitted. Thus, we can use

$$\tilde{\mathbf{P}}_1 = \mathbf{u}_1 \mathbf{v}_1^H. \tag{7}$$

We can interpret this as vector $\mathbf{v}_1^H$ is a single prototype pulse that has been extracted from the original pulse train and the vector $\mathbf{u}_1$ holds the complex amplitudes that are multiplied by the prototype pulse to create the pulses in the reconstructed pulse train.

3.4. Coding and compression ratio analysis

Summarizing, the above steps generate the following data that needs to be coded:

Side information:

1. The gating parameters T, G, and W
2. The missing pulse bit mask (if needed)
3. The $p-1$ time shifts for pulses in aligned pulse matrix

Signal information:

1. The p-element left singular vector (LSV) $\mathbf{u}_1$
2. The W-element right singular vector (RSV) $\mathbf{v}_1$

In this sub-section we provide a conservative assessment of the compression ratio achievable by the SVD-based compression. The analysis is conservative because (i) the number of bits allocated for coding data and side information is chosen on the conservative side and (ii) it assumes a somewhat worst-case scenario for dealing with missing pulse side information. Furthermore, to make the assessment more concrete we consider only the compression ratio relative to the size of the pulse matrix—that is, we take the pulse matrix as the "original data". This is done because we want to focus on the compressive effect of the SVD and we view the gating as something that would likely be done anyway as a crude means of compression. Thus, when computing the compression ratio the original data is taken as the pulse matrix together with the gating side information T, G, and W and the missing pulse bit mask.

The coding of T can be ignored because whether compression is used or not we must send the platform clock time of the first sample sent, which would be the time of the sample at index T. To code G would require no more than 32 bits to handle realistic pulse spacing at typical sampling rates; similarly, W could likely be coded with no more than 16 bits. Finally, the number of bits used to code each of the $p-1$ time shifts is $B_{TS}=8$ bits. The Appendix gives rationale for these choices; in fact, in most practical systems these parameters could be coded using fewer bits, but allocating an excessive number of bits to this side information has a negligible impact on the resulting compression ratio. As discussed in Section 3.1 we assume a $2p$-bit missing pulse mask (this assumes a missed pulse probability of 0.5). Thus, as shown in Table 1, the total side information for the compressed data will be conservatively assessed as $32+16+8(p-1)+2p=10p+40$ bits. The side information of the *original* data consists of T,

Table 1
Total compressed data with SVD coding of $p \times W$ aligned pulse matrix.

Quantity to code	General form of coding	Specific form of coding
Gating side information	B_G+B_W	32+16=48
Missing pulse bit mask	p/P_{mp}	$2p$
$(p-1)\times 1$ time shifts	$(p-1)\times B_{TS}$	$8(p-1)=8p-8$
$n\times 1$ RSV	$(W\times B_{RSV})$	$(8+1)W=9W$
$p\times 1$ LSV	$(p\times B_{LSV})$	$(8+4)p=12p$
Compressed data		$\mathbf{9W+22p+40}$

G, and W and the missing pulse bit mask, which will be $2p+48$ bits.

To code the signal data in the RSV we recognize that the RSV is a prototype of the pulse and therefore each of its samples is a complex number having magnitude and phase, both of which are changing from sample to sample. The cross-correlation processing will be much less sensitive to errors in the magnitude than in the phase, so we should ensure that the phase is coded with high accuracy whereas the magnitude can be coded with lower fidelity. We propose to code the phase of the prototype pulse using 8 bits/sample and to use a 1-bit differential PCM approach for the magnitudes of the prototype pulse. Thus, the number of bits used to code each element of the RSV is $B_{RSV}=8+1=9$ bits. It should be noted that this approach provides a fairly general approach that should work for virtually all cases; however, when the acquisition system identifies the radar as being a linear FM signal, the phase of the prototype should have fairly constant sample-to-sample phase differences, and then it may make more sense to use some form of differential coding there, too. To code the pulse magnitudes and phases contained in the LSV we again should allocate more bits to the phases than to the magnitudes. We use 4 bits per magnitude in the LSV and 8 bits per phase in the LSV. Thus, the number of bits used to code each element of the LSV is $B_{LSV}=8+4=12$ bits. The total number of bits used for the compressed signal data is also summarized in Table 1. The signal data part of the *original* data consists of $p\times W$ complex samples and we assume each complex sample is coded with 8 bits for its real part and 8 bits for its imaginary part.

How much compression can we get from this scheme? From above, the total number of bits for the original data is $16pW+2p+48$ and the total number of bits for the compressed data is $22p+9W+40$ so the compression ratio (CR) is

$$CR_{SVD}=\frac{16pW+2p+48}{22p+9W+40}. \tag{8}$$

Specific values of compression ratio results from Eq. (8) for typical practical scenarios are given in Table 2; the pulse width (PW), pulse repetition interval (PRI), and bandwidth (BW) values are for a few typical radars; the samples-per-pulse (W) values are dictated by the pulse width times a practical sampling rate with some extra samples counted per pulse to account for the fact that the gating can include some samples from outside the pulse;

Table 2
Compression ratios for typical practical scenarios.

PW (μs)	**BW** (MHz)	**# of pulses** p	**Samples/pulse** W	***CR***
0.5	4.0	80–300	6	4.3–4.4
1.5	2.0	1500–14,000	8	5.9–5.9
6.5	2.5	250–1500	24	16.9–17.4
9.0	2.8	60–400	38	22.0–26.7
49.0	4.5	80–200	300	85.4–154.2

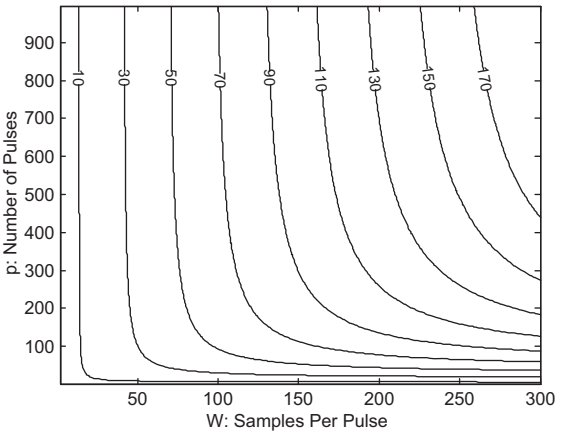


Fig. 7. Contour plot of Eq. (8) showing dependence of CR_{SVD} contours on the number of samples per pulse, W, and the number of pulses, p, collected.

the ranges of number-of-pulses is dictated by the need to achieve sufficient TDOA/FDOA accuracy under expected conditions. It should be noted that these compression ratio results are much lower than some preliminary results [26] because those earlier results did not consider the impact of coding the side information; nonetheless, even including the effect of the side information, as we have here, the compression ratios achieved are still very good.

A contour plot of the result in Eq. (8) is given in Fig. 7, which shows the dependence on the number of pulses p and the number of samples per pulse W. Large W occurs for long pulse width and/or high sampling rates (needed when pulses are phase/frequency modulated). Large p occurs when many pulses are collected to achieve the needed level of coherent processing gain within the cross-correlation. The largest compression ratios (several hundred to one) are achieved when both the number of pulses and the number of samples per pulse are large—these are precisely the scenarios where the largest amount of raw data is collected, so in a sense the compression ratio increases to manage the increasing amount of data that occurs as W and p grow. However, note that for a fixed small W the compression ratio grows very slowly with p; likewise, for a fixed small p the compression ratio grows very slowly with W. The typical region of operation in the W–p plane is shown in Fig. 7: W values below a few hundred and p values ranging from several 10 s up to several thousand. Note that in this typical region of operation, for a fixed type of radar (i.e., a fixed W value) the compression ratio is essentially constant as

p increases; thus, as we collect more pulses to achieve better coherent processing gain (and subsequently better accuracy on TDOA/FDOA) the improvement in compression ratio stagnates. This motivates the need for the pulse removal method in the next sub-section.

3.5. Removing pulses from the pulse matrix

Once the matrix is aligned but before performing the SVD-based compression it may be desirable to remove pulses (i.e., remove rows) from the pulse matrix. We have developed compression methods for general signals (i.e., not necessarily pulsed radar signals) that exploit a distortion criterion based on the Fisher information (FI) of TDOA/FDOA estimation [14–16]. The FI-based distortion criterion more completely captures the impact of compression on the TDOA/FDOA accuracies than does the classic MSE criterion that only captures the impact on signal fidelity. In previous work we have used the FI-based approach to allocate bits to wavelet packet coefficients in order to optimize compression for TDOA/FDOA accuracy [15,16]. Here we apply our FI-based distortion criterion in a different way: the idea is to remove entire pulses that have negligible contribution to FDOA accuracy by using FI to assess the contribution of each pulse to FDOA accuracy and then apply the SVD-based compression to a matrix with fewer rows.

As we have derived in [14], the FI measure for the data at the Tx platform for TDOA and FDOA are, respectively,

$$J_{TDOA} = \sum_{k=-N/2}^{N/2-1} \frac{2\pi^2 k^2 |\hat{S}[k]|^2}{N\sigma^2}, \tag{9}$$

and

$$J_{FDOA} = \sum_{k=-N/2}^{N/2-1} \frac{2\pi^2 k^2 |\hat{s}[k]|^2}{\sigma^2}, \tag{10}$$

where σ^2 is the noise variance of the data and $\hat{S}[k]$ is the DFT coefficient of the received signal data $\hat{s}[k]$. Note that the FI measure for TDOA depends on a measure of the frequency spread of the signal's DFT; thus, in principle, Eq. (9) could be exploited as follows. Taking the DFT of each pulse (i.e., across each row of the pulse matrix), take the magnitude squared of each DFT, and then average those down each column and use that result in place of $\hat{S}[k]$ in Eq. (9) to determine which columns of the DFT-processed matrix can be removed. However, we chose not to pursue this avenue because it is common to have fairly few samples per pulse and the frequency resolution would then be too poor to reliably apply Eq. (9). On the other hand, note that the FI measure for FDOA depends on a measure of the time spread of the signal itself and therefore removal of some specific pulse would reduce J_{FDOA}. Our goal here is to apply this idea and use Eq. (10) to determine which pulses will cause the smallest reduction in J_{FDOA}.

This insight motivates the following processing steps:

1. Choose the number of pulses, η, to be deleted (see below for guidance on how to choose η).
2. Compute $\Omega_i = \sum_{n \in i\text{th pulse index set}} n^2 |p_i[n]|^2$ for each pulse. (Note that the index n for the whole pulse train should run over a set that is symmetric around 0.)
3. Delete the η pulses having the smallest Ω_i.

To understand how to choose η, let $r = (p-\eta)/p$ be the ratio of the number of pulses left after pulse removal to the original number of pulses; so $0 < r \leq 1$. Thus, the compression ratio achieved by the pulse removal step is $1/r$. After removal of the pulses, the compression ratio achieved on the reduced pulse matrix by the SVD processing is found by modifying Eq. (8) by replacing p by rp and then accounting for the side information (SI in the equation below) needed to tell the decoder that which pulses were removed. Making these modifications gives

$$CR_{SVD}(r) = \frac{16rpW + 2pr + 48}{22rp + 9W + 40 + SI}; \tag{11}$$

a simple, conservative method for dealing with this side information is to send a p-bit mask with a one in the position of each retained pulse and a zero elsewhere. Then, using $SI=p$, the total compression ratio achieved by pulse removal followed by SVD processing is $CR_{SVD}(r)/r$, which gives

$$CR_{PR-SVD}(r) = \frac{16rpW + 2pr + 48}{22rp + 9W + 40 + p} \times \frac{1}{r}. \tag{12}$$

This function of r can be shown via a straightforward analysis[2] to monotonically increase as r goes from 1 to 0. Thus, the overall $CR_{PR-SVD}(r)$ improves monotonically as more pulses are removed. Of course, degradation in the accuracy of the estimates will also increase as more pulses are removed. Thus, for the given values of p and W and for a desired compression ratio value that exceeds that which is achievable without pulse removal, it is possible to determine from Eq. (12) the required r value and hence the required η value.

4. Simulation results

To illustrate the potential of SVD-based compression with and without FI-based pulse elimination we present some simulation results using a simulated radar signal (complex baseband) chosen to satisfy the assumptions made above. The signal is a train of pulses having linear FM modulation within each pulse. The simulated signal had the following parameters: 80 pulses, pulse width (PW) of 40 μs, a pulse repetition interval (PRI) of 70 μs, a maximum FM frequency deviation of ± 2 MHz, and a sampling rate of 4.92346 MHz. The PRI was set low for convenience to reduce the total number of samples used in the processing; this ensures that the time necessary to run simulations is not unreasonably long but has no effect on the compression since the gating removes any impact of the PRI on the compression ratio. The sampling rate was chosen so the sampling interval was incommensurate

[2] This can be re-written as $CR_{PR-SVD}(r) = a/(cr+d) + b/(r(cr+d))$ where $a=16pW+2p$, $b=48$, $c=22p$, and $d=9W+40+p$; clearly each term in this form $CR_{PR-SVD}(r)$ has the desired monotonic property because a, b, c, d are all positive.

with the PRI to ensure that the samples are not taken at exactly the same points within each pulse.

These radar specifications are shown in the last row of Table 2, where we see that the estimated CR using Eq. (8) for the case without pulse removal gives a range 85.4–154.2. Recall that Eq. (8) assumes the number of detected pulses is half of the total possible (i.e., P_{mp}=0.5); however, for the simulations we assumed that there are no missed pulses, so by removing the accounting of the $2p$-bit mask Eq. (8) becomes

$$CR_{SVD} = \frac{16pW+48}{20p+9W+40}. \tag{13}$$

In addition to examining the performance with only SVD-based compression we also consider the case of removing half of the pulses using the FI-based measure, so r=0.5 and Eq. (12) becomes

$$CR_{PR-SVD}(0.5) = 2 \times \frac{8pW+48}{10p+9W+40+p}, \tag{14}$$

where we have also not included the accounting for the missed pulse mask. For these conditions the compression ratio without pulse removal, from Eq. (13), is 88:1 and the compression ratio with pulse removal, from Eq. (14), increases to 106:1.

Monte Carlo simulations were performed to evaluate the impact of the compression on the standard deviation of the TDOA/FDOA estimation errors, where 400 runs were done for each evaluation. The first set of simulation runs was performed to establish the baseline performance of the SVD-based compression method without pulse elimination. The results are shown in Fig. 8, where the estimation accuracy is

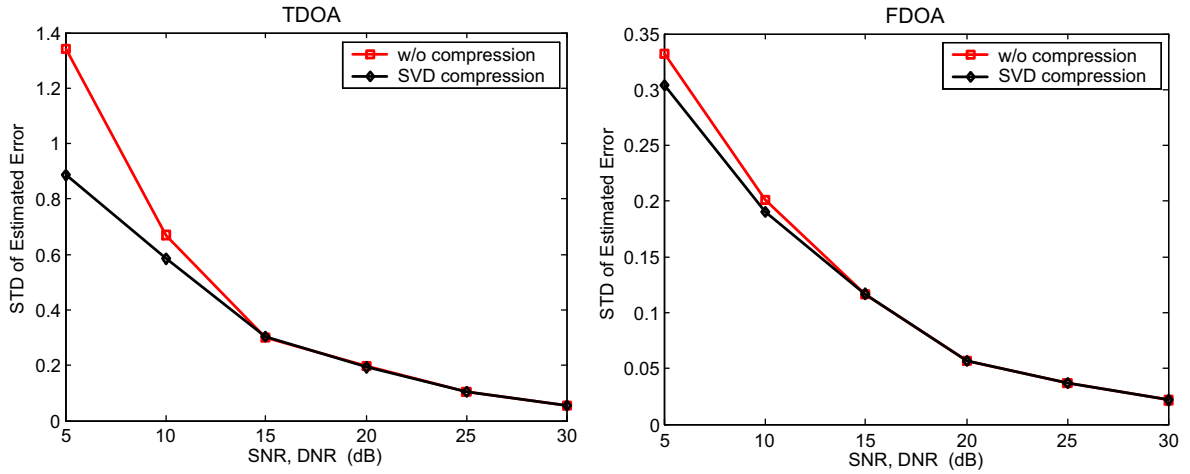


Fig. 8. Simulation results showing standard deviation of TDOA/FDOA estimation error for a simulated radar signal when compressed using SVD-based compression of the full pulse matrix (i.e., without pulse elimination) with CR_{SVD} of 88:1; in each plot the SNRs of the two cross-correlated signals are set equal to each other as they are varied over a range. For comparison, results are shown for the case of no compression.

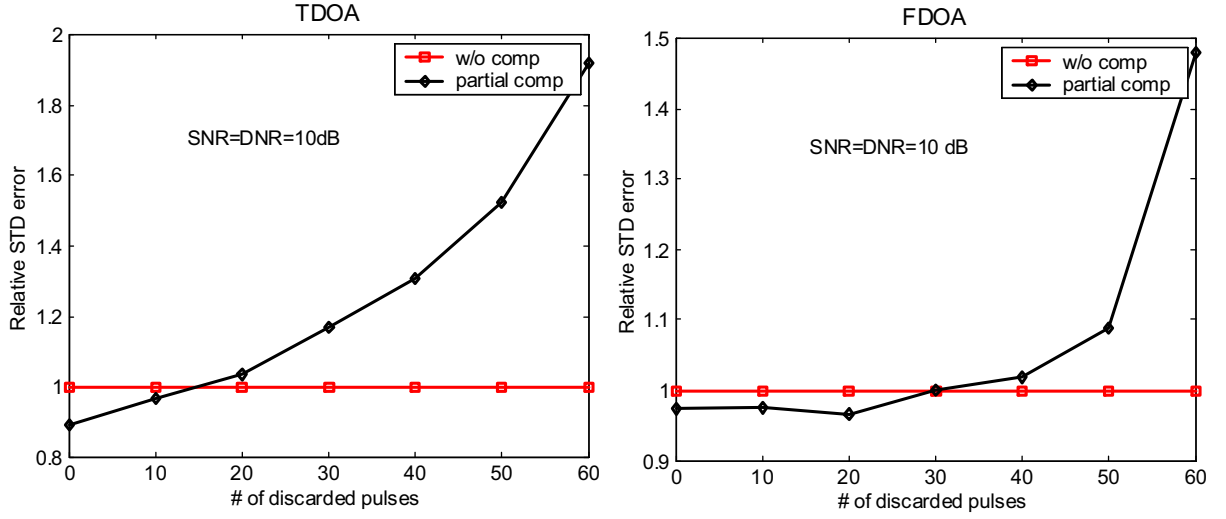


Fig. 9. Simulation results showing the effect that discarding pulses has on the TDOA/FDOA estimation error standard deviation. The term "partial comp" in the label means that SVD compression was done after FI-based pulse elimination; that is the SVD was applied to a "partial" pulse matrix,. For comparision, results are shown for the case without compression and pulse elimination.

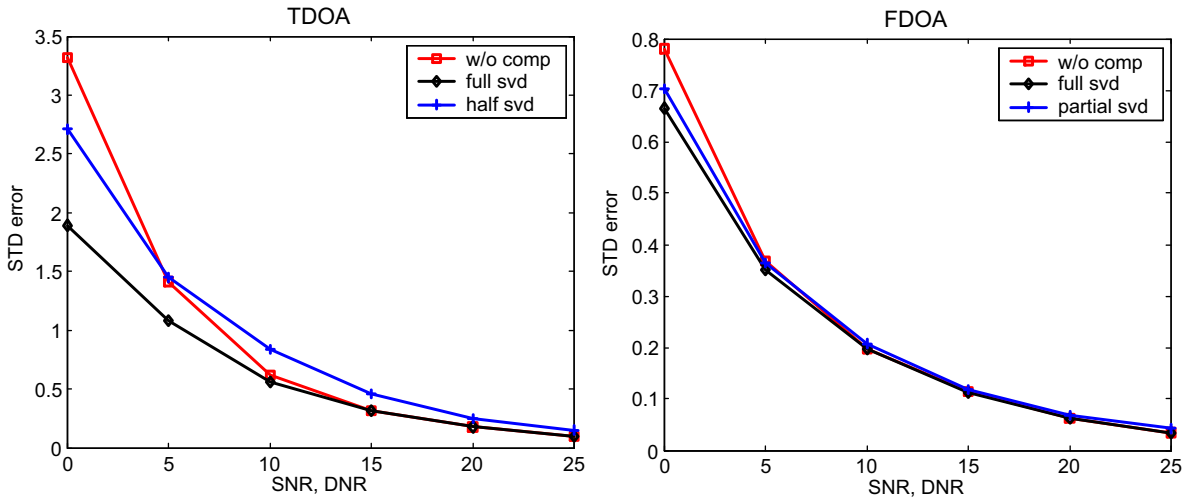


Fig. 10. Simulation results showing standard deviation of TDOA/FDOA estimation error for a simulated radar signal when compressed using SVD-based compression. The three curves compare the cases (i) without compression, (ii) full SVD-based compression (i.e., no pulse elimination) with compression ratio CR_{SVD} of 88:1, and (iii) SVD-based compression where half of the pulses are eliminated to give a compression ratio $CR_{PR-SVD}(0.5)$ of 106:1.

assessed as a function of *SNR* (with *SNR*=*DNR*) both with and without SVD-based compression. Surprisingly, even though the compression ratio is very high, at the lower SNR values the compressed signal actually yields *better* TDOA/FDOA accuracy than without compression. We believe that this is due to the well-known fact that the SVD provides an inherent de-noising property; however, surprisingly the de-noising seems to have less effect on the FDOA accuracy—an effect we currently do not understand.

As noted above, if pulses are eliminated according to the guidance of the FI then we expect negligible degradation in the FDOA accuracy up to a point; however, we would also expect the TDOA estimation error standard deviation to increase. Fortunately, the results in Fig. 8 indicate that, due to the de-noising, we have TDOA accuracy to "spare"; this, then, is a perfect setting for applying pulse elimination. To explore how many pulses can be eliminated, we ran simulations at the point *SNR*=*DNR*=10 dB and evaluated the error standard deviation after SVD compression with pulse elimination, as shown in Fig. 9, where the standard deviation values are presented relative to the value achieved without compression. For this case we see that we can eliminate half of the 80 pulses without suffering degradation in the FDOA accuracy, but there is some degradation of the TDOA when half of the pulses are eliminated.

Finally, for the case of eliminating half of the 80 pulses, the results in Fig. 10 show how the TDOA/FDOA accuracy varies with SNR. Through elimination of half the pulses the compression ratio increases from 88:1 to 106:1 but the TDOA/FDOA accuracy is, remarkably, still roughly the same as when no compression is used.

5. Conclusions

The basic idea of exploiting pulse-to-pulse redundancy in a radar pulse train for the purpose of data compression within TDOA/FDOA geo-location systems is quite simple in principle but requires several details to be carefully addressed to enable a viable algorithm. In this paper we have described how to combine several different signal processing ideas to address these issues and achieve very large compression ratios.

The key to the algorithm is the pulse matrix formed by placing a detected pulse in each row of the matrix. Time aligning the pulses in the rows of the pulse matrix is accomplished via correlation processing and fractional-delay processing. The resulting aligned matrix has only a single significant singular value and that is the key to the large compression ratios that have been achieved. Further compression is achieved by removing pulses that are deemed least significant to the FDOA accuracy on the basis of a distortion measure based on the Fisher information for FDOA estimation. The cost of the side information needed to be sent to the decoder was minimized via an optimization process applied to the gating of the pulses as they are placed into the matrix.

The result derived for the achievable compression ratio was shown to depend on the number of samples per pulse and the number of pulses collected. This relationship showed interesting and useful characteristics. First, the largest compression ratios are achieved when both the number of pulses (p) and the number of samples per pulse (W) are large, so the compression ratio increases to manage the increasing amount of data that occurs as W and p grow. Second, for a fixed small number of samples per pulse (a typical case) the compression ratio grows very slowly with the number of pulses. Thus, in many typical scenarios, as we collect more pulses to achieve better coherent processing gain (and subsequent accuracy on TDOA/FDOA) the improvement in compression ratio stagnates. This motivated the need for the pulse removal method that exploited the Fisher information based distortion measure, which was shown to allow

further removal of pulses while still maintaining quite good TDOA/FDOA quality.

All the compression ratio results were derived using conservative parameters for the coding of the post-SVD processing information. For example, the side information needed to code the gating parameters likely could be coded more efficiently—however, the gating side information is small compared to the coded SVD information and therefore using more efficient coding here would yield little improvement. Similarly, the compression ratio results were derived under conservative conditions on the number of pulses missed during the detection (which impacts the amount of side information needed). Despite the conservative conditions used for our compression ratio results, the predicted compression ratios are quite large in typical radar scenarios and enable rapid TDOA/FDOA-based geo-location of radar emitters that otherwise would be impossible to achieve over current and expected airborne data links.

Acknowledgement

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Appendix. Rationale for choice of number of bits for coding

To see why these bit values are feasible first recall that both G and W are integers because they are in units of samples. The integer G is bounded above by the largest expected PRI of the radars of interest, times the largest expected sampling rate; likewise, the integer W is bounded above by the largest expected PW, times the largest sampling rate. A typical value of a long PRI is 4 ms, so to bound the value of G we take an extreme value of 10 ms for PRI together with an extreme value of 10 Gs/s for the sampling rate, which gives an extreme bound on G of 10^8 which is much lower than $2^{32} \approx 4.3 \times 10^9$; thus, we can easily represent any realistic value of G using only 32 bits.

An unmodulated pulsed sinusoid has a mainlobe bandwidth of 2/PW, which puts a lower limit of 2 samples/pulse—although in practice on the order of 10 samples per pulse are used for an unmodulated pulse. For modulated pulses the number of samples per pulse must be increased, as it is driven by the sampling theorem. For modulated pulses the number of samples per pulse is on the order of a few tens of samples on the low side to a few hundred samples on the higher side; an extreme value would be a few thousand samples; certainly 65536 samples on a pulse is quite extreme. Thus, allowing 16 bits for the representation of W enables values from 0 to 65536 samples for the pulse and the extraneous samples inside the gate, which is more than enough.

The number of bits needed to code each time shift can be impacted by the quality of the front-end gating processing. The choice of 8 bits per time shift was determined to be acceptable via experimentation using a typical front-end quality.

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